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When Does Federated BatchNorm Help under Differential Privacy?

An Eval-Regime Characterization of DP-FedBN on Medical ECG Classification

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1. Introduction — The BatchNorm–Privacy Tension

- **Federated learning (FL)** trains models across hospitals without sharing patient data; **differential privacy (DP-SGD)** adds a formal guarantee to the updates that are shared.
- **FedBN** keeps each client's **BatchNorm (BN)** statistics local — a per-site domain adapter that is the key to non-IID medical data.
- But BN is **incompatible with DP-SGD**: per-sample gradients are ill-defined, and Opacus's ModuleValidator **refuses** any model containing BatchNorm.
- The standard fix — replace BN with **GroupNorm** — removes the very mechanism FedBN depends on.
- **Question:** can we keep FedBN's personalization **and** satisfy record-level DP — and when does it actually help?

2. Method — DP-FedBN (Dual-Optimizer)

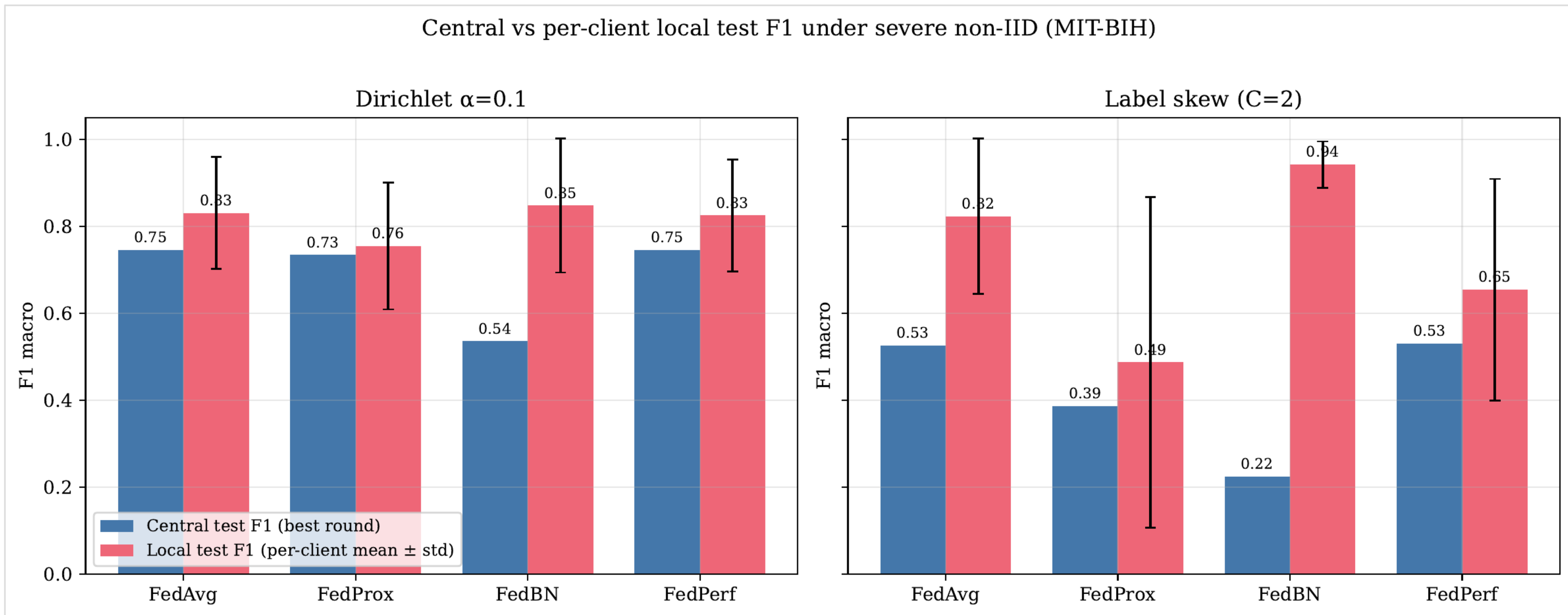
- **Key idea:** BN parameters and running statistics never cross the federation boundary, so they need **no DP**. Apply DP-SGD only to the parameters that are actually shared.
- **Per client, each round:**
  - **Freeze BN** → Opacus's ModuleValidator now accepts the model (freeze-then-unfreeze trick).
  - **opt_A** (AdamW + Opacus PrivacyEngine): non-BN params → per-sample **clipped + Gaussian-noised** (DP-SGD).
  - **opt_B** (plain AdamW): BN params → updated **locally, no DP**.
  - **Server:** FedAvg-average non-BN params; BN stays local (FedBN rule).
- Gives **record-level (ε, δ)-DP** on everything that leaves the client while preserving FedBN's per-client BN.
- First **record-level DP-SGD** composition of FedBN with explicit Opacus engineering (vs. the party-level Laplace "DP-FedBN" mention in ADCOL, ICML'23).

3. Experimental Setup

- **Datasets:** MIT-BIH Arrhythmia — 109,453 beats, 5-class AAMI, 360 Hz; PTB-XL — 21,388 records, binary Normal/Abnormal, Lead II, 100 Hz.
- **Model:** 4-block 1D-CNN, ~437k params, 5 BatchNorm modules. K = 5 clients, AdamW, batch 96.
- **Non-IID partitions (6):** IID, quantity skew, Dirichlet α ∈ {1.0, 0.5, 0.1}, label skew C = 2.
- **Privacy:** ε ∈ {1, 3, ∞}, δ = 1e-5, clip C = 1.0, Opacus 1.5. Methods: DP-FedAvg (raw BN), DP-FedAvg+GroupNorm, **DP-FedBN**.
- **Two eval regimes:** Central F1 (pooled holdout) vs. Local F1 (per-client). ~40 configurations, single seed (42).

4. Result 1 — The Eval-Regime Flip is Task-Agnostic

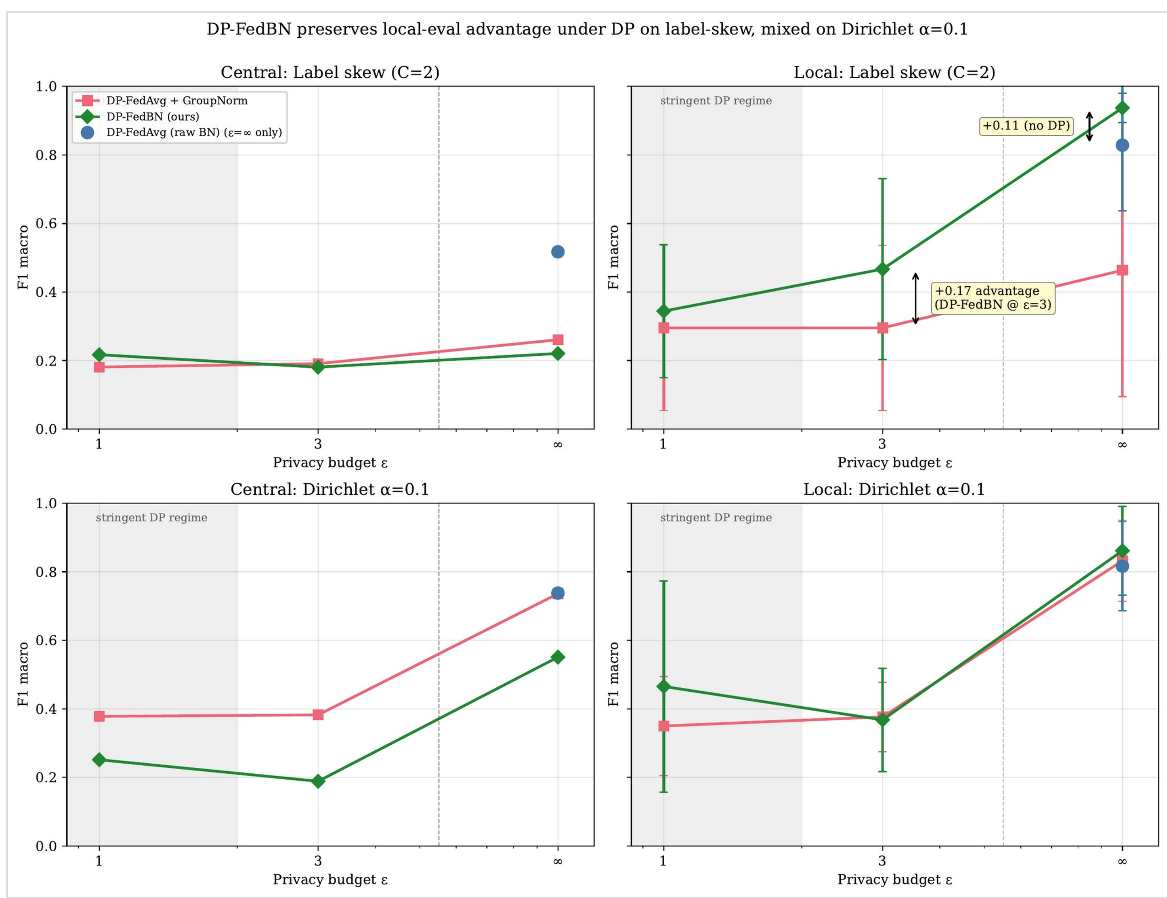
- On severe non-IID, FedBN ranks **worst on pooled-central F1** but **best on per-client local F1**.
- MIT-BIH label skew C=2: FedBN central **0.224** (lowest) vs. local **0.942** (highest) — a **+0.72 flip**.
- Replicates on PTB-XL (different task): FedBN−FedAvg = **−0.116 central / +0.123 local** — task-agnostic.



Central (blue) vs. per-client local (red) macro-F1 on MIT-BIH. FedBN: lowest central, highest local.

5. Result 2 — Under DP, the Edge is Partition-Dependent

- **Naive DP-FedAvg on a BN model is structurally impossible** — Opacus REFUSES it at every finite ε. This motivates DP-FedBN.
- **Label skew** (specialty-clinic regime): DP-FedBN keeps a **+0.17** local-F1 edge over GroupNorm at ε=3 (+0.05 at ε=1).
- **Dirichlet α=0.1** (proportional skew): edge vanishes/reverses (PTB-XL **−0.234** at ε=3) — GroupNorm wins. DP noise hits per-client BN with no cross-client averaging to smooth it.



Privacy–utility frontier (MIT-BIH). DP-FedBN (green) keeps a clear local edge on label-skew at ε=3; mixed on Dirichlet.

6. Conclusions & Practitioner Rubric

- The eval-regime flip is **real and task-agnostic** — always report **both** central and local F1.
- **Rubric:** label-truncated sites (specialty clinics) + local deployment → **DP-FedBN**; proportional case-mix (general hospitals) → **DP-FedAvg+GroupNorm**; strict pooled-central requirement → FedAvg/FedPerf, not FedBN.
- **Limitations:** single seed, 5 clients, ECG-only, fixed CNN. **Repro:** github.com/kanemoda/MedicalFL · Opacus 1.5 · PyTorch 2.4.